

EQUATIONS

THEORY

Equations

An equation is defined as a mathematical statement of equality.

Types of Equations

- a) Linear equation in one variable.
- b) Linear simultaneous equations in 2 or 3 variables.
- c) Quadratic equations.
- d) Cubic equations.
- e) Bi-quadratic equations.
- f) Exponential equations.

Quadratic Equations

- A quadratic equation is defined as polynomial equation of degree 2.
- A quadratic equation can be expressed in the following general form:

$$ax^2 + bx + c = 0; (a \neq 0)$$

- A quadratic equation can also be expressed in the factor form as follows:

$$a(x - \alpha)(x - \beta) = 0$$

Here, α and β are the roots or solutions of quadratic equations.

- The general solution of the quadratic equation can be obtained as follows:

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$\text{Sum of roots} = \alpha + \beta = -\frac{b}{a}$$

$$\text{Product of roots} = \alpha\beta = \frac{c}{a}$$

Structure of Quadratic Equations

If Sum (S) ($\alpha + \beta$) and Product (P) ($\alpha\beta$) of the roots are known, then the quadratic equation is

$$x^2 - Sx + P = 0$$

Sign of Roots of a Quadratic Equation

- When $c=0$, one root of the equation must be 0.
- When b and c are 0, then both the roots must be 0.
- If a, b, c all are of same sign, both roots are negative.
- If a and c are of same sign, opposite to that of b , then both the roots will be positive.
- If a and c are of opposite signs, one root is positive and another root is negative.

Nature of Roots

The expression " $b^2 - 4ac$ " is called the "Discriminant (D)" of the quadratic equation.

- When $D > 0$, Roots are real and distinct.
- When $D = 0$, Roots are real and equal.
- When $D < 0$, Roots are imaginary.
- When $D \geq 0$, Roots are real.
- When D is a perfect square, Roots are real, rational and unequal.
- When D is not a perfect square, Roots are real, irrational and unequal.
- If roots are equal use $b^2 = 4ac$.
- If roots are reciprocal of each other, use $a = c$
- If roots are equal but of opposite sign, use $b = 0$
- If roots are reciprocal but opposite in sign, use $c = -a$

Note

- Irrational roots will always appear in conjugate pairs.

$$\alpha = (a - \sqrt{b}) \text{ and } \beta = (a + \sqrt{b})$$

- Imaginary roots will always appear in conjugate pairs

$$\alpha = (a - ib) \text{ and } \beta = (a + ib)$$

Cubic Equations

- A cubic equation is a polynomial equation of degree 3, and the general form is represented as follows:

$$ax^3 + bx^2 + cx + d = 0; (a \neq 0)$$

- The factor form of a cubic equation is given as follows:

$$a(x - \alpha)(x - \beta)(x - \gamma) = 0$$

Here, α , β , and γ are the roots or solutions of the cubic equation.

- Sum of roots = $\alpha + \beta + \gamma = -b/a$
- Product of the roots = $\alpha\beta\gamma = -d/a$

Bi-Quadratic Equations

- A bi-quadratic equation is a polynomial of degree 4, and the general form is represented as follows:

$$ax^4 + bx^3 + cx^2 + dx + e = 0; (a \neq 0)$$

- The factor form of a cubic equation is given as follows:

$$(x - \alpha)(x - \beta)(x - \gamma)(x - \delta) = 0$$

Here, α , β , γ and δ are the roots or solutions of the bi-quadratic equation.

- Sum of roots = $\alpha + \beta + \gamma + \delta = -b/a$
- Product of the roots = $\alpha\beta\gamma\delta = e/a$